

1 2 Which quantity has a unit of kg m s^{-1} ?

[1 mark]

A force

☐

B impulse

☐

C moment

☐

D work done

☐

What is the relationship between the momentum p of a ball falling freely from rest and time t ?

[1 mark]

A p is constant

☐

B p is proportional to $t^{\frac{1}{2}}$

☐

C p is proportional to t

☐

D p is proportional to t^2

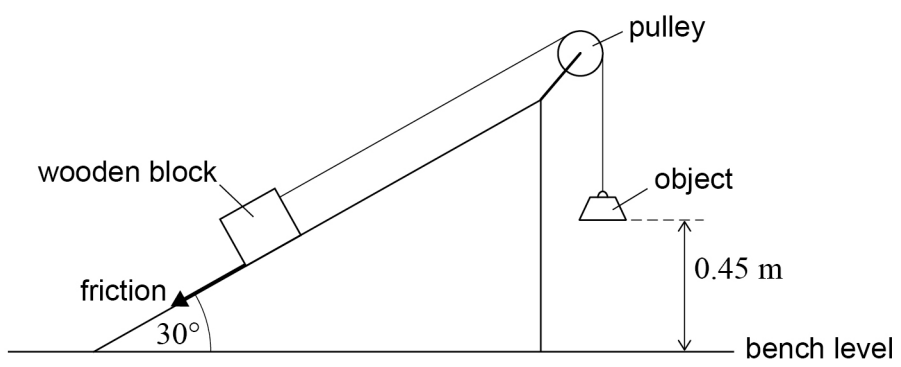
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Don't write

1 0

A student places a wooden block on a ramp inclined at 30° to the horizontal as shown in **Figure 7**. The block is pulled up the ramp by a light inextensible string connected, over a frictionless pulley, to an object of mass 0.40 kg .

Figure 7



The object is initially held at rest 0.45 m above the bench.
The wooden block has a mass of 0.25 kg and is initially at rest.

1 0 . 1

The object is released and accelerates vertically downwards at 2.23 m s^{-2} .

Explain **two** ways in which the momentum of the object is different from the momentum of the wooden block during this acceleration.

[2 marks]

- 1 _____

- 2 _____

1 0 . 2

Calculate the tension in the string as the object accelerates.

[2 marks]

tension = _____ N



1 0 . 3

Show that the object is travelling at approximately 1.4 m s^{-1} just before it hits the bench.

[2 marks]

1 0 . 4

Calculate the component of the weight of the wooden block that is parallel to the ramp.

[2 marks]

component = _____ N

1 0 . 5

When the object hits the bench, the string becomes slack and the wooden block decelerates and stops.
There is a constant frictional force of 1.25 N between the wooden block and the ramp.

Calculate the distance the wooden block travels parallel to the ramp as it slows down.

[3 marks]

distance = _____ m

END OF SECTION A

Section B

Answer **all** questions in this section.

*Do not write
outside the
box*

A rocket of mass 2.0×10^6 kg has an initial vertical thrust of 3.0×10^7 N at the Earth's surface.

What is the initial vertical acceleration of the rocket?

[1 mark]

A 5.2 m s^{-2}

☐

B 14 m s^{-2}

☐

C 15 m s^{-2}

☐

D 25 m s^{-2}

☐

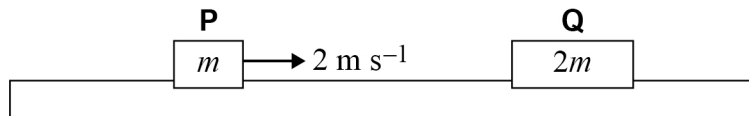
1 8

Trolleys **P** and **Q** are on a frictionless horizontal runway.

P has an initial velocity of 2 m s^{-1} towards **Q**.

Q is initially stationary.

P has a mass of m and **Q** has a mass of $2m$.



P collides with **Q**. Immediately after the collision **Q** has a velocity of $+\frac{4}{3} \text{ m s}^{-1}$.

What is the velocity of **P** immediately after the collision?

[1 mark]

A $+\frac{2}{3} \text{ m s}^{-1}$

B $+\frac{1}{3} \text{ m s}^{-1}$

C $-\frac{1}{3} \text{ m s}^{-1}$

D $-\frac{2}{3} \text{ m s}^{-1}$



Do not write

A car is in collision with a truck. The car experiences an impulse Δp .

When is the impulse experienced by the truck equal to $-\Delta p$?

[1 mark]

A Only when the collision is elastic.

☐

B Only when the collision is inelastic.

☐

C Only when the truck and the car have the same initial momentum.

☐

D Whenever no external forces act during the collision.

☐

2	1
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A ball travels at speed u before colliding with a wall.
During the collision it loses 20% of its kinetic energy.

What is the speed of the ball immediately after the collision?

[1 mark]

A $0.89u$



B $0.80u$



C $0.64u$



D $0.36u$



Turn over ►



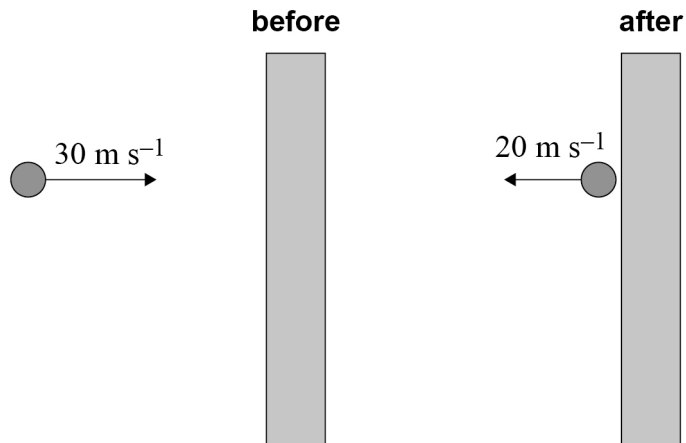
Do not write

1

7

outside the
box

A ball of mass 0.1 kg hits a wall and rebounds.



What is the magnitude of the impulse that acts on the ball?

[1 mark]

A 1 N s

☐

B 2 N s

☐

C 3 N s

☐

D 5 N s

☐

07

A falcon is a bird of prey that catches other small birds in flight. The falcon dives to increase its speed. Its gravitational potential energy decreases by 1900 J during the dive.

The mass of the falcon is 1.1 kg.

07.1

Calculate the vertical distance moved by the falcon as it dives.

[2 marks]

vertical distance = _____ m

07.2

The initial vertical speed of the falcon is 25 m s^{-1} .

During the dive, the falcon's wings apply a vertical force downwards, transferring 1400 J of work.

Calculate the maximum theoretical speed that the falcon could achieve in this dive.

[3 marks]

maximum speed = _____ m s^{-1}

07.3

In practice, the falcon will need to transfer more than 1400 J of work to achieve this maximum theoretical speed.

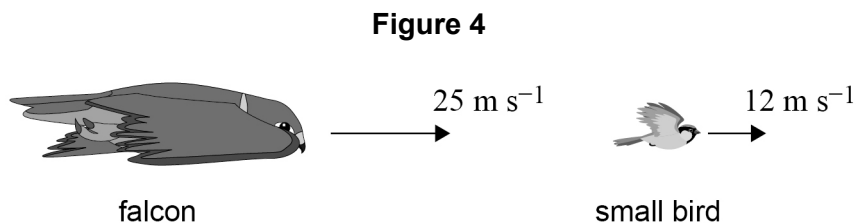
Explain why.

[2 marks]



On another occasion, the falcon catches a small bird while gliding horizontally.

Figure 4 shows the falcon and the small bird gliding horizontally in the same direction at different constant speeds.



The falcon collides with the small bird and grips it tightly.

The mass of the small bird is 0.18 kg.

The initial speed of the falcon is 25 m s^{-1} .

The initial speed of the small bird is 12 m s^{-1} .

0 7 . 4

State the law of conservation of momentum.

[1 mark]

0 7 . 5

Calculate the speed of the falcon immediately after this collision.

[3 marks]

speed = _____ m s^{-1}



Do not write Do not write

Which row gives the fundamental (base) unit for the quantity indicated?

[1 mark]

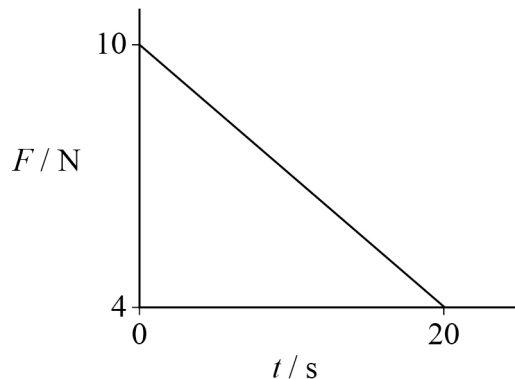
	Quantity	Fundamental (base) unit	
A	energy	kg m s^{-2}	☐
B	force	kg m s^{-1}	☐
C	moment	$\text{kg m}^2 \text{s}^{-2}$	☐
D	momentum	$\text{kg m}^2 \text{s}^{-2}$	☐

Turn over ►



Do not write Do not write

The graph shows the variation of force F with time t for an object accelerating from rest.



What is the magnitude of the momentum of the object after 20 s?

[1 mark]

A 140 N s

☐

B 120 N s

☐

C 60 N s

☐

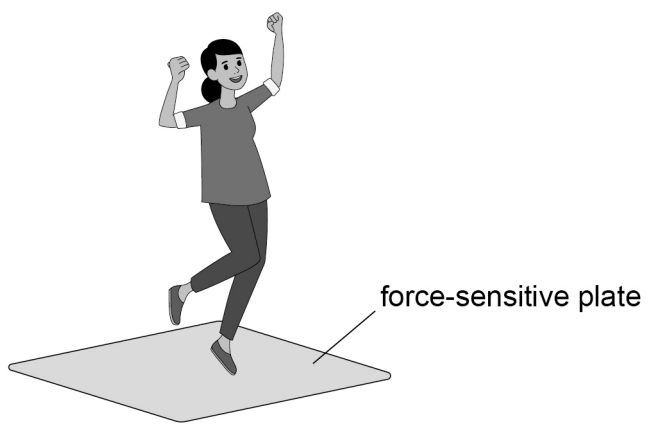
D 0.30 N s

☐

0 8

A student stands on a force-sensitive plate and then jumps upwards and lands as shown in **Figure 8**.

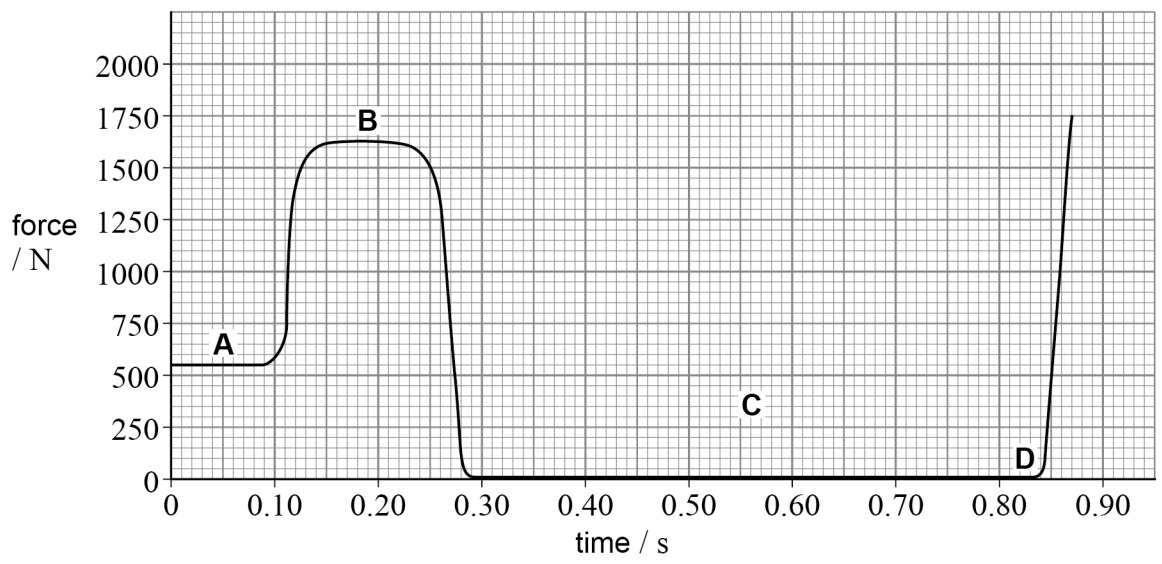
Figure 8



A data logger attached to the plate measures the variation with time of the force exerted by the student on the plate.

Figure 9 shows the variation of force with time as the student stands (**A**), jumps up (**B**), is in the air (**C**) and begins to land (**D**).

Figure 9



0 8

1

Show that the mass of the student is approximately 56 kg.

[1 mark]



08.2

Show that the student gains approximately 150 kg m s^{-1} of momentum when she jumps up. Use the part of the graph labelled **B**.

[3 marks]

08.3

Calculate the speed of the student as she leaves the plate.
Use your answer to Question **08.2**.

[1 mark]

speed = _____ m s^{-1}

08.4

Calculate the time the student is in the air.
Use your answer to Question **08.3** and a suitable equation of motion.

[2 marks]

time = _____ s

08.5

Determine, using section **C** of **Figure 9**, the time the student is in the air, as recorded by the data logger.

[1 mark]

time = _____ s

Turn over ►



08.6

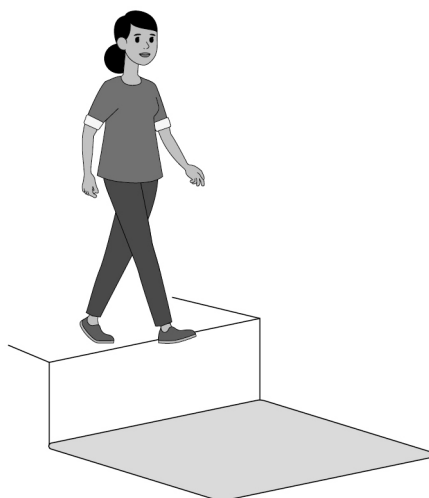
Explain, without further calculation, whether your answer to Question **08.4** or your answer to Question **08.5** has the smaller uncertainty.

[1 mark]

08.7

Next, the student steps off a platform onto the force-sensitive plate, as shown in **Figure 10**.

Figure 10



The student steps off the platform twice:

- the first time that she lands, she keeps her legs straight
- the second time that she lands, she bends her knees, increasing the time taken to decelerate.

Explain how increasing the time taken to decelerate affects the force on the plate. Refer to one of Newton's laws of motion in your answer.

[3 marks]

END OF SECTION A



Section B

Answer **all** questions in this section.

*Do not write
outside the
box*

1 **4** Which row contains two scalar quantities?

[1 mark]

A	velocity	mass	☐
B	speed	distance	☐
C	weight	momentum	☐
D	time	acceleration	☐

Turn over ►



Do not write Do not write

An apple falls from a tree through a vertical height of 3 m.

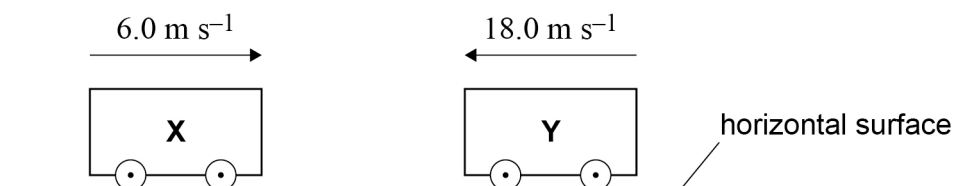
Estimate the momentum of the apple just before it hits the ground.
Give your answer to an appropriate number of significant figures.

[3 marks]

momentum = _____ kg m s⁻¹

Figure 1 shows two trucks **X** and **Y** that are moving towards each other on a horizontal surface.

Figure 1



X has a mass of 4.8 kg and is moving to the right at a speed of 6.0 m s^{-1} .
Y has a mass of 2.4 kg and is moving to the left at a speed of 18.0 m s^{-1} .

The two trucks collide.

Show that truck **X** changes direction during the collision.

[3 marks]

Do not write

1 7

A spacecraft of constant mass is initially at rest.
 A constant resultant force F acts on the spacecraft.
 The change in momentum of the spacecraft is Δp .
 The change in kinetic energy of the spacecraft is ΔE_k .

Which row shows the relationship between Δp and F and between ΔE_k and F ?

[1 mark]

A	$\Delta p \propto F$	$\Delta E_k \propto F$	☐
B	$\Delta p \propto F$	$\Delta E_k \propto F^2$	☐
C	$\Delta p \propto F^2$	$\Delta E_k \propto F$	☐
D	$\Delta p \propto F^2$	$\Delta E_k \propto F^2$	☐

1 6 Trolley **P** has a mass m and moves with a velocity v on a horizontal surface.
P collides with a stationary trolley **Q**.
Q has a mass m . The collision is elastic.

What are the velocities of **P** and **Q** immediately after the collision?

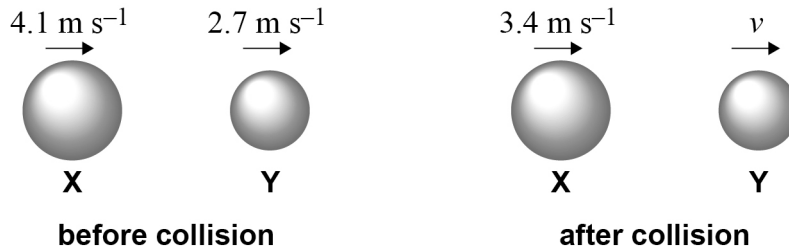
[1 mark]

	Velocity of P	Velocity of Q	
A	$\frac{v}{2}$	$\frac{v}{2}$	☐
B	$\frac{3v}{2}$	$\frac{3v}{2}$	☐
C	$-v$	0	☐
D	0	v	☐

Figure 4 shows two objects **X** and **Y** immediately before and immediately after they collide.

The velocities of **X** and **Y** before and after the collision are shown on **Figure 4**.

Figure 4



The mass of **X** is 8.4 kg and the mass of **Y** is 6.3 kg.
 The total kinetic energy immediately before the collision is 94 J.

Determine the velocity v of **Y** immediately after the collision.
 Go on to determine whether the collision is elastic or inelastic.

[4 marks]

$v = \underline{\hspace{4cm}} \text{ m s}^{-1}$



Don't write

1**3**

Which shows only vectors?

[1 mark]

A acceleration, displacement, work

☐

B acceleration, distance, force

☐

C displacement, force, momentum

☐

D force, kinetic energy, mass

☐

1	4
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A ball of mass 2.7 g is released from a height of 0.30 m above a surface. It bounces on the surface to reach a height of 0.23 m.

What is the change in momentum of the ball at the surface?

[1 mark]

A $8.2 \times 10^{-4} \text{ kg m s}^{-1}$

☐

B $5.7 \times 10^{-3} \text{ kg m s}^{-1}$

☐

C $6.6 \times 10^{-3} \text{ kg m s}^{-1}$

☐

D $1.2 \times 10^{-2} \text{ kg m s}^{-1}$

☐

Don't write

2**3**

The area under a force–time graph gives the change in:

[1 mark]

A displacement.

☐

B momentum.

☐

C velocity.

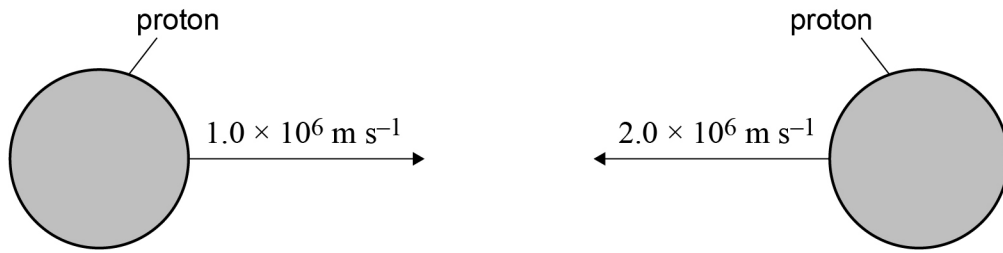
☐

D work done.

☐

Figure 1 shows two protons travelling towards each other just before they collide. The speeds of the protons are $1.0 \times 10^6 \text{ m s}^{-1}$ and $2.0 \times 10^6 \text{ m s}^{-1}$.

Figure 1



Deduce whether the kinetic energy of this collision can produce an electron–positron pair.

rest energy of an electron = 0.511 MeV

[3 marks]

3

Turn over for the next question

Turn over ►



Don't write

1 7 Which row is true for an inelastic collision?

[1 mark]

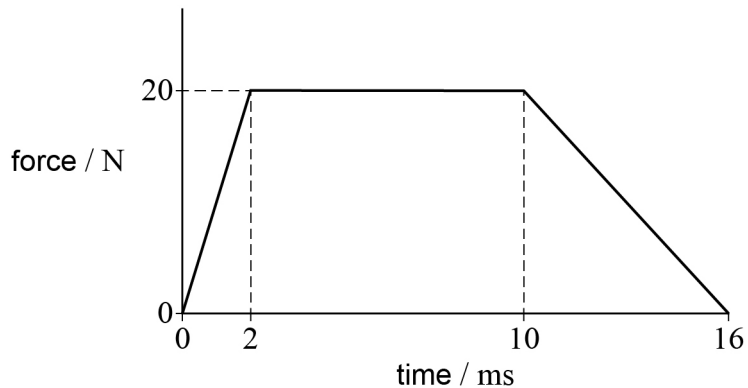
	Momentum	Kinetic energy	
A	conserved	not conserved	☐
B	conserved	conserved	☐
C	not conserved	conserved	☐
D	not conserved	not conserved	☐



Don't write

1 8

A force is applied to a ball. The graph shows the variation of force with time.



What is the change in momentum of the ball between 0 and 16 ms?

[1 mark]

A $240 \times 10^{-3} \text{ N s}$

☐

B $320 \times 10^{-3} \text{ N s}$

☐

C 240 N s

☐

D 320 N s

☐